Randolph Hall • Anthony Boccanfuso
Editors

University-Industry Collaboration

Innovation at the Interface

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Editors
Randolph Hall
Epstein Department of Industrial
and Systems Engineering
University of Southern California
Los Angeles, CA, USA

Anthony Boccanfuso
University-Industry Demonstration
Partnership (UIDP)
Columbia, SC, USA

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Chapter 1
Alignment, Engagement, and Public Benefits

Anthony Boccanfuso and Randolph Hall

1.1 Introduction

The Nobel Prizes awarded in October of 2024 were stunning for how they brought traditional sciences—chemistry and physics in particular—into the world of computation and artificial intelligence. For physics, the award recognized foundational research on neural networks. For chemistry, the award recognized the transformational software AlphaFold, which predicts protein structures. As noted in *The Economist* (Uncredited, 2024) “In both cases, the research being awarded would be seen by a stickler as being outside the remit of the prize-giving committees.” In recognizing neural networks and software like AlphaFold, the Nobel committees have demonstrated the transformative shift toward an increasingly interdisciplinary fields of science.

The Physics and Chemistry Nobels, as well as the Nobel Prize for Physiology and Medicine, were notable in another way: all had leveraged University-Industry (UI) relationships. Physics Prize winner Geoffrey Hinton, from University of Toronto, spent a decade at Google. Chemistry Prize winners Dennis Hassabis and John Jumper developed AlphaFold through their work at Google’s Deepmind. All three, along with collaborators, developed tools that enabled scientific discovery in universities. Whereas the Physiology and Medicine winners had not worked in industry, their research on MicroRNA has been transformational for biomedical

A. Boccanfuso
University-Industry Demonstration Partnership (UIDP), Columbia, SC, USA
e-mail: tony@uidp.net

R. Hall (✉)
Epstein Department of Industrial and Systems Engineering, University of Southern California, Los Angeles, CA, USA
e-mail: rwhall@usc.edu

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companies, through use in therapeutic development, applicable to broad classes of disease.

As these examples illustrate, innovation today often occurs at the “interface,” the interstitial space where industry and universities work with each other toward common goals. Innovation at the interface, as our book is titled, can occur throughout the chain of science, engineering and scholarship, along the way bridging theory, experiments, discovery, test, development, design, implementation and commercialization.

Fundamental to our book is the notion that strength and impact are developed not solely through relationships between universities and industry but *engagement*, in research and scholarship, education and learning, and public benefits. Engagement entails a purposeful and aligned collaboration, usually formalized through agreements. UI engagement is a distinct feature of the American university system that emerged after the Civil War, a consequence of the Morrill Land-Grant Act as well as the nation’s appetite for industrialization and efficiency. Since then, universities and industry have found ways to support each other, not just in the U.S.A. but throughout the world. In recent decades, the pace of innovation at the interface has accelerated, producing a multitude of rewards, described as follows.

Research and Scholarship As demonstrated in recent Nobel Prizes, collaboration can catalyze the creation of knowledge, contextualizing science and discovery, creating instruments and software for data generation and analysis, conducting trials and experiments in real-world settings, and engaging human subjects in studies. UI collaboration can broaden the experiences of faculty and students, satisfying their curiosity for discovery and helping turn their ideas into technologies that benefit humankind. Participants in innovation at the interface can achieve notoriety through citation of their research, publicity, honors and introduction into the membership of national academies, such as the National Academies of Science, Engineering and Medicine (NASEM), and the National Academy of Inventors (NAI).

Education and Learning Universities prepare students for careers via their degree programs, which are sometimes tailored to specific types of jobs that industry is seeking to fill. The UI connection in education is particularly strong in professional programs, such as masters degrees in engineering, and also in continuing and executive education, in business and administration, targeting working professionals. Industry enriches degree programs through class lectures, projects, cooperative education, internships, and participation in graduate and undergraduate dissertations. Universities and industry work together to upskill existing workers, making curriculum relevant to evolving needs and developing a workforce that matches needs for particular skills and capabilities.

Public Benefit UI collaborations contribute to communities and nations in ways that increase well-being. They can grow economies and increase wealth as startup companies are founded from university-generated inventions, technology commercialized by established corporations, and alumni contribute to businesses. UI collaborations can produce innovations and insights that lead to new medical technologies and pharmaceuticals, reductions in pollution or increased productivity or help solve "grand challenges" (e.g., climate change and extending lives) facing society (Brint, 2018, summarizes numerous achievements in his Chap. 3).

Research and scholarship, education and learning, and public benefit speak to the core missions of universities, what University of California President Clark Kerr has called the "multi-university" (Kerr, 2001). The multi-university is broad and expansive, touching communities and society in diverse ways, as centers for culture, education, discovery, innovation and even health care. However, to succeed in these three areas, universities must do two other things: secure financial resources and earn a positive reputation. Financially, UI engagement generates university income through sponsored research, royalties from licensed technology, service payments, gifts and corporate memberships. Though funding like this helps pay for the costs of UI collaboration, it should not be the driver for engagement. In the United States, businesses typically support less than a 6% slice of total academic research expenditures. Even the universities with exceptionally large industry funded research programs (more than $150 million/year, at University of Texas M.D. Anderson, Duke, University of Pennsylvania, and MIT), top out at 20% of their totals (NSF, 2023). The reality is that the federal government is by far the largest sponsor of research in universities, supporting well over half. While universities benefit from income received from industry, it constitutes a small portion of university budgets.

With respect to reputation, universities care deeply about how they are perceived by their stakeholders, peers, sponsors, and the public at large. Their notoriety is reflected in media, rankings, expressed sentiments, and statistics, each of which may place the university in a positive (or negative) light, compared to others or standing on their own. Universities also care about feedback loops, how an elevated reputation can catalyze a virtuous cycle, attracting more and stronger students, outstanding staff and faculty, donations, government funding and other resources that can be used to improve the institution (Fig. 1.1). It also matters to universities how a drop in reputation—lower rank, negative stories, bad experiences—can catalyze a downward spiral, leading to fewer resources.

From our experience, industry has similar goals as universities, but in different proportions. In his classic New York Times article, "The Social Responsibility of Business is to Increase Its Profits," Milton Friedman (1970) argues that the corporate executive is responsible to conduct the business in accordance with the desires

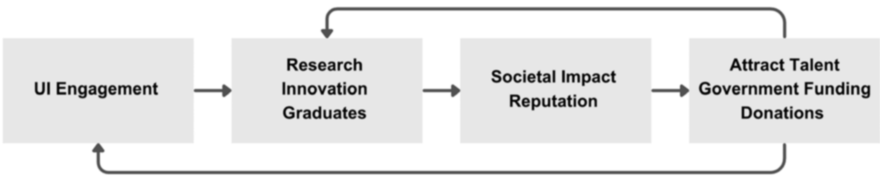

Fig. 1.1 Feedback loop whereby UI engagement elevates reputation and attracts talent and resources

of the owners, "which generally will be to make as much money as possible while conforming to the basic rules of the society, both those embodied in law and those embodied in ethical custom." Profitability is the consequence of income exceeding costs, tracked over time and considering expected returns on investment.

According to Friedman, public benefits and expansion of knowledge may be good things, but should matter for the corporate executive only as an instrument toward the result of profitability. A counter point-of-view comes from the Business Roundtable (2019), which created a list of social, community, human capital, and environmental outcomes—along with profitability—within its redefined "Business Purpose of a Corporation." Similar melding of public and private goals can be found within Environmental, Social and Governance (ESG) investments and Social Purpose and Benefit Corporations, which explicitly allow boards to consider social impacts of investments in addition to profitability.

To summarize, both universities and companies have financial goals. Though companies pursue profits, which may then be distributed to owners and investors, universities seek to cover their costs of operation and investments, including strategic investments. Both universities and industry also care about their reputation, though they may be looking for different aspects of reputation, with universities more explicitly seeking notoriety as an end result rather than an instrument toward profitability. Figure 1.2 shows how shared aims drive UI partnerships. Research and scholarship, education and learning and public benefit matter to both universities and companies, but need to be viewed in the context of "toward what end" and in "what proportion." Alignment is clear toward the goals of workforce development,

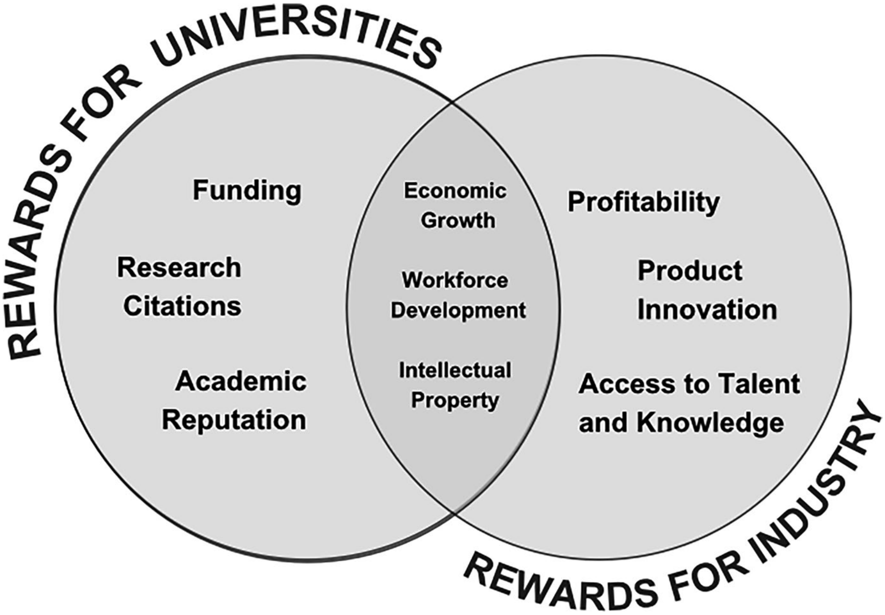

Fig. 1.2 Rewards, shared interests, and alignment

economic growth, and intellectual property, but more distinct when it comes to profitability and academic reputation. The pathways from a research discovery to the ends of money and reputation are quite different for companies and universities, with the former more immediately seeking monetization through product innovation and the latter more immediately seeking research impact through citations. Nevertheless, universities are quite cognizant that reputation will in the end attract gifts, sponsored research, licensing income, and other types of funding. Whereas getting known for a great accomplishment—particularly prominent prizes like the Nobel—can be greatly satisfying for universities, companies cannot sidetrack profitability.

1.2 Opportunities and Challenges

Successful UI engagement depends on participants aligning their differing goals and practices toward common purpose, considering how intermediate outcomes, like inventions, knowledge, and workforce development, might eventually elevate their reputations and financial health. We begin with examples of success, and we then introduce the challenges to overcome, reflecting the competing objectives that can obstruct success.

1.2.1 Examples of UI Engagement Success

The chapters that follow offer many stories of how UI engagement produced innovations that solve society's problems. For example:

IBM and Stanford Chapter 5 describes how research on catalysis that does not need heavy metals like platinum was jump started through collaboration. A family of organic molecules that were effective in catalyzing polymerization and depolymerization reactions was created, leading to discovering a way to depolymerize Polyethylene Terephthalate (PET), offering a way to recycle plastic bottles and polyester fabric used increasingly in performance clothing.

Boeing and University of Sheffield Chapter 6 offers the Advanced Manufacturing Research Centre (AMRC) in Sheffield, UK, as a case study. The AMRC was conceptualized in 1999 by The University of Sheffield in a proposal to research hard metals machining (titanium and super alloy steels). The expertise of the university combined with a multi-year commitment by Boeing broke free a significant UK government grant to stand up AMRC's first 10,000 square foot building, fill it with state-of-the-art equipment and hire a full time research staff. The initial success enabled the AMRC to recruit new industry members, each bringing a multi-year funding commitment. As of 2024, the AMRC employed 600+ researchers, technicians and support staff and engaged over 120 industry members in a 40 million GBP annual research portfolio.

Houston Methodist (HM) and Medtronic Chapter 10 shows how "Applied Medical Innovation" (AMI) can accelerate innovation for medical devices within academic health centers. Medtronic partnered with the Houston Methodist's DeBakey Heart & Vascular Center and other clinical sites to conduct clinical trials on the repositionable, self-expanding, transcatheter aortic-valve replacement (TAVR). TAVR changed the paradigm in how to treat valve repair. Without the partnership between industry (Medtronic) and academic health centers, the innovation would not have been possible.

Mars, University of California at Davis and University of Arkansas Chapter 17 tells how Mars, UC Davis and the University of Arkansas partnered to study a specific rice watering technique called "alternate wetting and drying" (AWD). AWD involves flooding a rice field and allowing the water to naturally subside to the soil's surface. This climate-smart agricultural method significantly reduces water use and greenhouse gas emissions with no yield reductions. AWD can also be used with other conservation practices to reduce water and energy consumption, improving farmers' profits.

As these examples demonstrate, when companies and universities partner toward mutual goals, the participants are not the only beneficiaries. The public at large benefits from the development and adoption of advanced technology for health, the environment, efficiency, and safety, as well as from economic development in innovation centers. Nevertheless, success is not easy, as we describe in the next section.

1.2.2 Challenges

Companies and universities have strikingly different histories and models of governance. The most highly ranked universities in America are all more than a century old, whereas the largest and most dominant companies were all founded in the last 50 years. Universities use "shared governance" between faculty and administration (who may also be faculty), and operate as not-for-profit corporations under boards of trustees. Companies are for-profit and those of size are typically publicly traded. Newer startups are typically privately held with significant ownership for founders and venture investors. A very important difference is how universities value academic freedom, meaning faculty and students have significant autonomy to pursue their personal passions, with few constraints as to how they might share the fruits of their work with others. By design, universities are "porous," permitting people, ideas, and products to flow in and out, whereas companies constrain movement of valued IP—inventions, data, software, publications, and media. The difference can be attributed to the relative importance of producing work that elevates reputation versus the importance of producing work that elevates commercialization.

UI engagement can be easy at inception, but challenging in execution. Because universities are "porous," company representatives can easily access faculty and students to learn about ongoing research and studies. Further, enormous amounts of information is publicly available through university websites, published papers, symposia, open data sets and software, and repositories. It is not hard for a company to learn what universities are doing and find potential collaborators because faculty and students are motivated and incentivized to share their achievements.

The challenge for UI engagement is taking the next step from simple meetings to true engagement, especially when money is paid for execution of work or acquisition of products. At that point relationships are defined through agreements, such as sponsored research, technology licenses, services, or consulting. Policies, regulations, laws, and ethics surrounding agreements are discussed in depth in Chap. 14. But as a preview, these are some examples of the things companies and universities must agree on:

- Does the industry sponsor have rights to review or approve research prior to its publication?
- How must IP resulting from research be protected, who pays the costs, and what IP rights are granted to the sponsor?
- How are the rights of students working on a project protected?
- What types of deliverables or products must be provided as a condition for payment; what is the schedule for deliverables; how do they align with the needs of all parties (e.g., timelines for student dissertations, class schedules, pace of competition, and need to demonstrate results)?
- What are the charges for the direct cost of completing work as well as the indirect costs (administration and facilities); how are these funds used within the university?
- How are the rights of individual companies protected when multiple companies are sponsoring similar projects in the same laboratory, or companies participate as a consortium?
- What are the obligations of faculty, staff, and students to disclose their personal relationships with companies; what rules are in place to project the objectivity of the research; what rules are in place to assure their loyalty to the institution?
- What rights are conveyed for use, ownership and protection of data, and how might it be shared with others?
- Is it simpler and more effective to fund an engagement through an irrevocable gift rather than a contract with specified deliverables?

Within universities, each item can result in a time-consuming negotiation among multiple parties, including faculty principal investigators, individual schools or departments, the department of contracts and grants and the technology licensing office. Similarly, with companies, multiple parties may participate in negotiation, including research partners, management, contracting and so on. Some negotiations are never completed because of hard-line differences in goals, or because the expectations of one party are impossible to guarantee on the part of the other (e.g., when strong assurances are needed for the actions of independent-minded faculty, or when a company intends to use research as a vehicle to promote a particular point of view or a particular product).

The faculty leading research projects as principal investigators must carefully consider how an industry project fulfills their academic goals. Perhaps it catalyzes new work that could not be conducted in the absence of collaboration. It would be a mistake to assume UI research projects are "applied," though they might be. A UI project may be the inspiration for new ideas that would not otherwise have been conceived, as in the work co-editor Hall led in cooperation with the United Parcel Service on models for territory planning, blending computer algorithms with the knowledge acquired by humans through their work.

Some authors have criticized universities for sacrificing their values in the pursuit of lucrative industry relationships. In *University, Inc.*, Washburn (2005) wrote "The emergence of a utilitarian, market-model university, combined with a loud drumbeat calling on schools to spur national and regional economic growth, now threatens to obliterate the distinctiveness of …academic research culture." On the other hand, as said at the introduction to this chapter, novel research might be impossible without industry contributions, due to their unique skills and infrastructure, and their potential as a living laboratory for observing and testing ideas.

One more challenge needs to be mentioned. As noted in co-editor Hall's (2024) book *Managing Innovation Inside Universities*, universities and industry today often compete with each other. In contrast, during universities' formative years between the Civil War and World War I, they did not compete, as the leading companies produced physical products—food, fuel, steel, and manufactured goods—none of which are produced by universities at scale (then or now).

Today, leading companies, like universities, produce, communicate, process, store, and analyze information. Companies, like universities, are capable of delivering educational courses and programs. They manage enormous libraries derived from books, articles, and other resources. They hire large teams of PhD-trained researchers to develop insights from the data they possess. Universities and industry have become more similar to each other as enterprises that thrive through reputation and income derived from information.

Ultimately, effective UI engagement depends on matching complementary talents and capabilities in ways that benefit all participants, within a culture founded on trust and respecting commitments. We will return to an important mechanism for doing just that in a later section of this chapter, when we describe UIDP in depth.

1.3 Prior Research

To put our book in context, we now describe prior scholarship, introducing books and other resources already published that synthesize university history, innovation ecosystems, entrepreneurship, and UI engagement. Later, in Chap. 2, our book will provide a more complete review of important studies on the topic. We also note that Rybnicek and Konigsgruber (2019) offer a systematic review of literature on UI engagement, as a reference.

1.3.1 Historical Perspectives

American universities can trace their origin as far back as the seventeenth century, when Harvard was founded. But they did not emerge as true universities until the period after the Civil War. Veysey (1965) traces the transition of America's small colleges into comprehensive universities, incorporating graduate education, research and scholarship, the emergence of Johns Hopkins as the nation's first doctoral degree granting institution, and creation of the nation's system of land-grant universities, supported in the Morrill Act of 1862. The model established then provided the framework for how universities would engage with industry to today. The Act required supported universities that would "promote the liberal and practical education of the industrial classes in the several pursuits and professions in life." The Morrill Act accomplished several things. First, higher education was broadened from what had been a type of finishing school for the elite to practical education for many more people. Second, professional education became an explicit goal, leading to universities that emphasize engineering and agriculture, in support of America's emerging industries. Last, the Act established that public higher education would be led by individual states, with federal support.

Five decades after passage of the Morrill Act, the Smith Lever Act of 1914 took one more step, establishing agricultural extension services at land-grant universities for applied research, education and training in agriculture, which continue to this day. Extension services use federal funding to directly support the agricultural industry in America, developing and improving crops by bridging research and practical application.

Key features of America's private and public universities were well established by 1920:

- Doctoral education through which students acquire the skills of research through mentored dissertations (inspired by German universities).
- Professional degree programs through which students would be trained in industry practices.
- Engagement and interaction with industry through professional programs, while serving national goals toward economic development.
- Freedom and objectivity as institutional values, applicable to faculty and students, and institutions as a whole.

Unlike other nations, the U.S.A. never adopted a national university system, leaving public universities to the states and simultaneously providing avenues for independent private universities to thrive. According to Hollingsworth and Hollingsworth (2011, p. 27), "There is substantial literature which suggests that continuous high levels of radical innovation in modern societies require diversity in organization forms, heterogeneity in organizational structures, diversity in ideas," all features of the American system. In total, America created a unique place for universities, through which freedom and objectivity inspired innovation within universities, while catalyzing practical application for public benefit.

Various authors attribute America's economic advancement over the last century to its unique form of higher education. Important books include *Higher Education in America*, written by former Harvard President Derek Bok (2013), *The Great American University*, written by former Columbia Provost Jonathon Cole (2009), *Creation of the Future*, written by former Cornell President Frank Rhodes (2001), and *The Uses of the University*, written by former University of California President Clark Kerr.

1.3.2 Innovation Ecosystems

In *How Innovation Works*, Matt Ridley (2020, p. 6) writes "Innovation happens when people are free to think, experiment and speculate." He also writes that "for innovation to flourish it is vital to have an economy that encourages or at least allows outsiders, challengers and disruptors to get a foothold." Ridley's principles, built on the concepts of the economist Joseph Schumpeter (1942), offer a foundation for innovation "ecosystems," which can be characterized by networks of people and organizations that innovate through their mutual interactions. An innovation ecosystem can be place based—as in Silicon Valley and the San Francisco Bay Area—or can be global, such as the ecosystem surrounding research methods, such as AlphaFold.

A key insight on ecosystems is that even for-profit corporations benefit from a flow of ideas and people, at least within limits. This may include employees changing employers, company acquisitions and mergers, and licensing of technologies. It could also include more organic forms of idea flow, as in conferences, alumni networks or social encounters by happenstance. Put another way, innovation does not readily occur in isolation.

Colleges and universities offer anchors within regions from which innovation ecosystems emerge—Stanford for Silicon Valley and MIT for Boston, as examples. The creative interactions in research universities, in particular, lead to new technologies and companies, which may later spawn others. Fischman et al. (2014) and Roberts and Eesley (2009) examined MIT's innovation strategy in depth, emphasizing how freedom catalyzed creativity among students, faculty, and alumni.

As noted by Scott and Kirst (2017) in *Higher Education and Silicon Valley*, place-based ecosystems also need a range of colleges and universities, not just one elite university. In California, under its Master Plan for Higher Education, the educational ecosystem includes community colleges, California State Universities and the University of California. They write that "As the seat of a world-famous innovation economy, this region's success rests upon knowledge and training of its workforce," by which they mean the entirety of the workforce, not just the founders of companies. However, the significance of the University of California cannot be understated, as nowhere else in the world are there so many highly ranked universities as California, including two in the San Francisco Bay Area (UC San Francisco and UC Berkeley).

Prominent companies and research laboratories (both private and government) can also be anchors. Silicon Valley emerged after World War II not just from Stanford, but from the companies nearby, along with government investment, as told in O'Mara's *The Code* (2019). She writes, "success came thanks to a vibrant and diverse cast of thousands, not just the marquee players" in a "place filled with smart people who had mostly come from somewhere else."

In the post WWII era, government investment in research and development was inspired by work under the Manhattan Project (led by J. Robert Oppenheimer, a UC Berkeley faculty member) and radar technology from MIT's Radiation Laboratory. Though research in these two areas was secretive and classified, Vannevar Bush's deeply influential report to the U.S. President—*Science, The Endless Frontier* (1945)—emphasized the strategic importance of investment in fundamental and open basic research (a theme emphasized again in the 2007 NASEM report *Rising Above the Gathering Storm*). One consequence was the creation of the National Science Foundation. Another was support, within the Department of Defense, for fundamental research programs at universities, which complemented restricted research at companies and national laboratories. Etzkowitz and Leydesdorff (1995) characterize the feedback loop among government, universities, and industry as a mutually supportive "triple Helix," a phenomenon documented in O'Mara's *The Code*.

Other regions have subsequently sought to emulate Silicon Valley, building economies that thrive through UI engagement. Since the great recession of 2008, state and local governments have looked to universities as catalysts for economic development, supporting both industry and workers. In recognition of this goal, the Association of Public and Land Grant Universities has established the Innovation and Economic Prosperity Program, which offers universities a designation for success in regional economic engagement, growth and economic opportunity (APLU, 2021).

Various other authors have examined the relationship between universities and economic development in depth. Important sources include Thorp and Goldstein (2013) in *Engines of Innovation*, Geiger and Sa (2008) in *Tapping the Riches of Science*, and Lane and Johnstone (2012) in *Universities and Colleges as Economic Drivers*.

1.3.3 Entrepreneurship and Technology Transfer

Technology transfer and entrepreneurship is another important aspect of UI engagement. In some cases, engagement begins after inventions, software or other Intellectual Property (IP) has been conceived. Other times, industry research sponsorship may result in new IP. Under the Bayh-Dole Act of 1980, IP created from U.S. government funding at universities is owned by the university, with rights conveyed back to the government, obligations for how IP is managed and an obligation to share IP income with inventors. U.S. university policies for IP comply with federal requirements. They also typically require the creators of IP to assign IP resulting from nonfederal sponsorship to the university. On the other hand, IP resulting from the work students do in classes is typically owned by the creators. IP regulations vary among nations. In some cases, individual universities are free to set their own policies. In Sweden, IP is owned by inventors, not the university, by law.

With regard to UI engagement, corporations negotiate rights to IP resulting from the research that they sponsor with university contracts and grants offices. They then negotiate licenses to IP with a university technology transfer office. Both of these offices tend to fall under a Vice President of Research for the university. Sometimes they are merged together, perhaps within an office of corporate engagement.

In these processes, a challenge for universities is resolving conflicting goals, particularly whether the best pathway for commercialization is a startup company, for which the creator may be a founder, or an established corporate entity, for which the creator may have no ownership stake. The university may want to support the entrepreneurial spirit of its faculty or may want startup companies as a step toward economic development; however, most startups fail, established companies can invest in further technology development, and startups can create unresolvable conflicts of interest. Further, by virtue of a sponsored research agreement, universities may well be obligated to give an existing company first rights to the IP.

Several books comprehensively examine these questions, including *Building Technology Transfer Within Research Universities* (Allen & O'Shea, 2014), *Building Effective Technology Transfer Offices* (Cunningham et al., 2020), *University Technology Transfer, What it is and How to Do It* (Hockaday, 2020), and, more broadly, *Managing the Research University* (Smith, 2011).

1.3.4 UI Engagement

We turn last to the focus of our book, UI engagement, aspects of which can be found in the books already cited in this chapter. The following chapter also provides a comprehensive review of the literature from historical and international perspectives.

There are comparatively few books focused on UI engagement, of which we provide three examples. *Strategic Industry-University Partnerships* (Frolund & Riedel, 2018) offers numerous case studies, all from the perspective of individual companies, such as IBM, BMW, and Novo Nordisk. The cases speak to the motivations behind strategic investments made by companies, along with success factors and metrics, such as access to talent, IP, and shared research. Developing *University-Industry Relations* (Miller & Le Boeuf, 2009) provides a brief summary of key UI engagement topics, along with guidance in how a university might integrate various offices into a "one-stop shop" to interface with industry. More recently, *University-industry Partnerships for Positive Change* examines UI engagement in the context of broader social purposes, specifically toward addressing the United Nations' Strategic Goals (Bodley-Scott & Oymak, 2022). The book emphasizes complementary capabilities of universities and industry, and how alliances can become transformative, rather than just transactional.

1.3.5 Summary of What Is Known About UI Engagement

Prior research offers insight into how universities have integrated UI engagement into their missions of research and scholarship, education and learning and public service, beginning in America during the formative years after the Civil War. It has also shown that universities have anchored place-based innovation ecosystems (e.g., Silicon Valley and Boston). We know from prior research how universities and companies have been organized to facilitate engagement, as well as how they seek to overcome some of the obstacles. We next turn to the value of collaborative communities, where industry and university representatives can meet to discuss and surmount the challenges, attaining alignment, engagement, and public benefits. UIDP shows what is possible when university and industry work together to build trusted engagements.

1.4 University Industry Demonstration Partnership

A central challenge in UI engagement is execution of contracts and other agreements between the companies that sponsor research and the universities that perform the work. These challenges were a foundational element in the University Industry Demonstration Partnership's (UIDP's) origin story.

There have been countless contemporary efforts to streamline and simplify the contracting process, all of which play a role in the organization's history. In the 1980s, the National Academies' Government-University-Industry Research Roundtable (GUIRR) was established. GUIRR has a history of organizing efforts aimed at streamlining contracting, particularly for federally sponsored research. For instance, it became the convener for the Federal Demonstration Partnership (FDP).

GUIRR began an initiative focused on UI relations in 2003, producing a model agreement for negotiating IP. However, it was immediately clear that a one-size-fits-all approach would not work. At the same time, other stakeholders and organizations wrangled with the topic, including the Industrial Research Institute (IRI) and the National Council of University Research Administrators (NCURA), among others. Cross-sector committees produced white papers with valuable contributions but no real breakthroughs. Key stakeholders Merrilea Mayo from GUIRR, Bob Killoren (Ohio State University) from NCURA, and Susan Butts from Dow's External Technology Group realized more was needed. They understood that the problem required an objective third-party committee to transition the accumulated learnings toward practitioners. The new committee would expand beyond contracting templates by creating a forum to facilitate multilateral understanding, improve communication, and support human connections that produce breakthroughs in UI collaboration.

The committee produced a white paper in 2004, proposing the creation of the University-Industry Demonstration Partnership (UIDP) as an activity of the National Academies. The committee later crafted a set of founding principles, published in 2006 as the *Guiding Principles to University-Industry Endeavors*. The group received initial funding from key parties, including the National Science Foundation, the Kauffman Foundation, UCLA, the University of Illinois, Hewlett Packard, Pfizer and ExOne, and invited interested companies and U.S. universities to join.

An early work product of UIDP, called TurboNegotiator, was the basis for its Contract Accords. Interestingly, TurboNegotiator was 20 years ahead of its time and, in many ways, was the original machine learning tool to help with contract negotiation (a number of nascent tools are now available and being used by universities with modest levels of success).

The Contract Accords was crafted to address specific contract clauses that generated the most challenges when negotiating an agreement. It has continued to grow and evolve since the original work. Originally addressing five specific contract clauses, the accords now address nearly 20 separate items, providing overviews, academic and corporate perspectives, and strategies to resolve differences. This work would become seminal for moving research partnerships forward and establishing UIDP as the authority in UI collaboration.

While the Contract Accords were the initial product, there was a strong desire to consider other important issues that hindered stronger UI partnerships. Prior to 2009, a team of Hewlett Packard executives, led by Wayne Johnson, developed a graph showing the different modes of engagement that the company had with universities as they progressed through different phases of their UI relationship. This graph served as inspiration for the 2012 publication of the UIDP Partnership Continuum. This publication brought awareness to the multitude of ways that companies and universities can work together. One of the underlying assumptions of the original Guiding Principles and Partnership Continuum was that all UI relationships should over time strive to advance from transactional relationships to strategic alliances. This is no longer the case. Many companies are instead identifying a small strategic set of partner organizations for holistic engagement.

While continuing to develop additional modules for the Contract Accords, UIDP used the Partnership Continuum work to extend beyond contracting issues into other areas affecting partnerships—talent, strategic relationships, and innovation ecosystems and economic development.

1.4.1 Project Process

Soon after its creation, the UIDP created a project process that leveraged the lessons learned from the development of the Contract Accords. The process uses the deep experiences of subject matter experts to identify issues and develop practical solutions.

All project ideas originate with members and reflect university and industry needs related to contracting, research administration, strategic partnerships, innovation ecosystems, and talent pipeline topics. Through various curated crowdsourcing techniques, projects leverage UIDP's robust network of experts and practitioners with various job roles related to UI collaboration and fostering impactful cross-sector partnerships in research and innovation.

Working groups are used to divide and conquer tasks. They are also a great way for members to network with each other. In most cases, UIDP hires an expert to compile a draft prior to input of the working group. The expert will use the information approved by the Project Committee (composed of board members, university reps, and industry reps) and their knowledge of the content area. Once the draft is nearly final, the working group provides feedback. UIDP uses these steps to ensure members are not overburdened.

After an idea is identified, a brief project application, based on the "Heilmeier Catechism," is filled out. The Project Committee provides feedback and approves the project. The Project Committee also reviews the final draft and votes it into the review process. The process identifies people who were not involved in the writing of the document, who review it for completeness, accuracy, and clarity. Project outcomes are carefully vetted through a modified peer-review and approval process to ensure alignment with generally accepted principles of the UIDP community. Then, the Project Committee votes on the approval of the final product. The product is then socialized through numerous channels to members. Once complete, UIDP member organizations apply the guidance from UIDP resources to their day jobs and their organizations' approaches.

1.4.2 Example Projects

UIDP has dramatically expanded its project portfolio beyond the Contract Accords. As of the release of this book, there are over 60 “how to” guides/publications that provide unique insights into ways to strengthen UI partnerships. Some examples follow.

Collaboration Metrics A menu of more than 100 metrics that companies and universities can use to evaluate their UI partnerships, plus how to use the metrics. For many companies and universities, identifying compelling metrics that effectively convey the value (e.g., return on investment), and quality of UI collaborations has been exceedingly difficult. This resource enables companies and universities to select measurable, purpose-driven indicators for evaluating their partnerships and determining which ones to grow.

Co-Investment by Companies and National Science Foundation (NSF) Multiple projects have focused on co-investment by government and companies in fundamental, use-inspired research, including a Joint Solicitations Quick Guide (focused on co-investment in university research) and Joint Solicitations Comparison Tables (detailed comparison of all the features of existing joint solicitations). UIDP played an integral role in developing the most recent NSF joint solicitation, MSF-SPEED for Molecular Foundations for Sustainability, supported by UIDP members Procter & Gamble, PepsiCo, BASF, Dow, and IBM.

Insights for Researchers A compilation of researcher-focused guidance from over 15 different UIDP publications. This resource is designed to help researchers understand the points of view of their university and industry counterparts on topics ranging from collaborator identification to budgeting and IP.

Partnership Continuum & Avenues for Engagement Since its inception, UIDP has been identifying partnership modalities and detailing different collaboration models. These projects are just two of many that raise awareness of the myriad ways a company and university can collaborate.

Academic Research Engagement Teams This project provides an overview of different reporting structures companies may use when strategically working with universities. The project’s guide delves into the formation of an “Academic Research Engagement Team (ARET),” addressing where an ARET may be situated within a company, how an ARET is staffed and funded, and different types of university programs that an ARET may be involved in.

1.4.3 Standalone Organization

After more than a decade under the auspices of the National Academies, the UIDP had matured to exist as a standalone organization. In late 2014, the UIDP Board came to an agreement with the National Academies to graduate, as of July 2015, and become an independent, nonprofit organization. Since UIDP industry members work globally, they expressed interest in expanding UIDP's university membership beyond the U.S. and, in 2016, UIDP launched a pilot program by extending invitations to key non-U.S. universities, including Oxford, Toronto and Tokyo. This pilot yielded significant benefits to the membership. The experiment validated the rationale for growing the non-U.S. university membership.

UIDP membership is by invitation and includes universities from around the world as well as multinational companies who possess the capabilities to have strong academic collaborations. Smaller organizations benefit from UIDP's work by accessing many of the program's learnings (many of which are publicly available) and participating in activities.

While government agencies are not extended invitations to join, many have participated in UIDP activities over the years and invested in its work through non-recurring awards in areas of the partnership's expertise. More recently, UIDP has engaged private, philanthropic organizations and funders that are interested in partnering with companies and universities to create new discoveries that can be translated to new products and services and improve the human condition. One example is The Aligning Interests in Support of Chemistry Research workshop, funded by NSF, which resulted in the MFS-SPEED solicitation that supports The National Defense Authorization Act (NDAA) for Fiscal Year 2021 and the 2022 CHIPS and Science Act. Another example is "Building Multinational Teams to Address Global Challenges," which featured representatives from the National Institute of Standards and Technology, Pew Charitable Trust, NSF, and the Structural Genomics Consortium. This workshop utilized the University of Cambridge road mapping framework to drive discussions on facilitating cross-sector, cross-border collaborations to better the human condition.

UIDP's international reputation has grown. The organization has been invited to host numerous events at locations outside the U.S. and now 12% of its membership is non-U.S. organizations. UIDP's impact is further demonstrated by its continued growth and its unique university-to-industry ratio, unmatched by any similar organization. Additionally, UIDP's membership is elite, with both university and industry members continuously ranked at the top of their respective fields.

1.5 Takeaways and Future Strategies

UI engagement is a way that universities and industry "innovate at the interface." To succeed, they must overcome the challenges and barriers to collaboration, aligning goals for mutual benefit within formalized relationships, elevating reputation and financial viability for both parties. UI engagement occurs across multiple dimensions, most importantly research and scholarship, education and learning, and public service.

Our book draws from experts on key aspects of UI engagement, many of whom are active contributors to UIDP. Chapter authors include practitioners of UI engagement—from universities and from companies—who have drawn from their deep experience as to what has worked, and not worked, in their own universities and companies. Chapter authors also include academicians who have studied the evolution and effectiveness of UI engagement. They provide perspective as to the options that have been used over time, why the solutions have changed and where UI engagement is heading in the future. The authors' diverse perspectives and deep experience give readers an understanding of the motivations, strategies, and tactics to design and implement successful UI programs.

University-Industry Collaboration is divided into three sections, addressing the foundations, avenues, and mechanics of UI engagement (Table 1.1):

Foundations (Chaps. 1–5) addresses the evolution, organizational models, motivations, and success metrics for UI engagement. In addition to a broad international overview, specific examples of UI engagement are provided for Princeton University and IBM Corporation.

Avenues (Chaps. 6–13) addresses the areas in which UI engagement is making a difference, including strategic alliances and consortia, innovation centers and economic development, entrepreneurship, executive education, academic health systems, and clinical trials. The section concludes with strategies for gaining support for UI engagement, first within companies and second within universities.

Mechanics (Chaps. 14–17) addresses the specific steps needed to create and sustain UI relationships, including ethical and legal frameworks, intellectual property, research data, and last international relationships.

Table 1.1 Chapter topics: foundations, avenues, and mechanics

Foundations	Avenues	Mechanics

• Evolution of UI engagement
• Organizational models
• Motivations and success metrics
• International comparisons
• Examples of engagement from:
  • IBM
  • Princeton
	• Strategic partnerships and consortia
	• Innovation centers and economic development
	• Entrepreneurship
	• Executive education
	• Academic health systems
	• Clinical trials
	• Gaining support within:
	  • Companies
	  • Universities
		• Ethical and legal frameworks
		• Intellectual property management
		• Research data management
		• International relationships

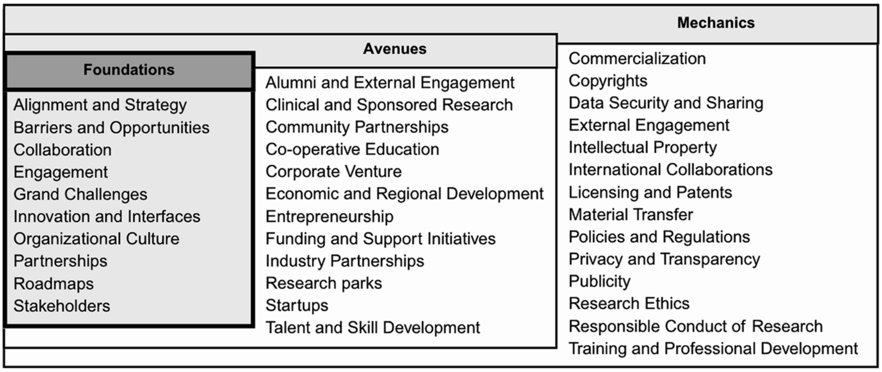

Fig. 1.3 Key issues in UI collaborations: foundations, avenues, and mechanics

Collectively, *University-Industry Collaboration* explores the why and how of UI engagement in all of its dimensions. We offer a guide for those who seek impactful research and innovation that integrates the talents and strengths of students, faculty, researchers, and developers. We draw from the framework of Fig. 1.3 by addressing the foundations, avenues, and mechanics of catalyzing, initiating, and sustaining impactful UI engagements. Our theme is alignment—understanding and connecting the goals of partnering organizations toward common purpose. From alignment comes the strength of partnership for "innovation at the interface" of universities and industry.

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Anthony Boccanfuso Since 2007, Anthony Boccanfuso is President and CEO of UIDP, a solutions-oriented global membership organization composed of top-tier innovation companies and world-class research universities. UIDP supports mutually beneficial collaborations by developing and disseminating strategies for addressing common issues among the sectors—academic, corporate, government, and nonprofits. After completing his doctorate in inorganic chemistry, Boccanfuso embarked on a career spanning more than 30 years in research and commercialization roles, managing a variety of administrative, programmatic, and strategy initiatives for academic, government, and private sector organizations. Boccanfuso's wife Laura is an accomplished entrepreneur. They have three grown children, Carolina, Michael, and Ana Catherine.

Randolph Hall is Dean's Professor of Industrial and Systems Engineering and Director of CREATE at the University of Southern California. His experience includes serving for nearly 15 years as USC's VP of Research, where he oversaw research advancement, administration, and ethics for the entire university. He is the author of *Managing Innovation Inside Universities* and *Queueing Methods for Services and Manufacturing*, and Editor of the *Handbook of Transportation Science, Patient Flow,* and the *Handbook of Healthcare System Scheduling*. He has written numerous articles on how universities can innovate in their practices and previously served as board chair for the UIDP. Hall holds a BS in Industrial Engineering and Operations Research and a PhD in Civil Engineering from UC Berkeley.